

Giovanni SAMBO

ABOUT ME

I am a committed student who always tries to face new challenges. The many experiences made in the sports and professional field have helped me to relate to multiple realities that have defined my salient qualities as leadership, adaptability, time management, motivation and team working. I am keen on the field of aerospace engineering. In particular, concerned with the development of innovative technologies which might positively impact the whole society and environment.

CONTACT DETAILS

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in @GiovanniSambo

SKILLS

Programming languages:

- MATLAB (advanced)
- C (basic)

Aero & CFD:

- SU2
- OpenFOAM
- Paraview & Gmesh
- Wind tunnel testing

3D modelling:

- SolidWorks
- Inventor
- Solid Edge

Others:

- XFOIL
- Dewesoft
- NASA-CEA
- Excel

LANGUAGES

Mother tongue: Italian

Others: English, Spanish

INTERESTS:

- Aircraft
- Renewable energy
- Travelling (22 country visited)
- Sports
- Hand-builder RPAS

EDUCATION

Master Aeronautical Engineering

Politecnico di Milano, Milan, Italy

September 2026

- ◊ **AERODYNAMICS & FLUID DYNAMICS:** incompressible and compressible flows, with focus on experimental techniques, CFD applications and hands-on wind tunnel testing.
- ◊ **WIND& ENERGY SYSTEMS:** advanced study of wind-structure interaction and atmospheric boundary layer flows. Focus on the aerodynamic and structural design of wind turbines for renewable energy applications.
- ◊ **FLIGHT PERFORMANCE, FLIGHT TESTING & PROPULSION**

Erasmus Exchange Programme

ETSi, Sevilla, Spain

25/01/2024-20/06/2024

- ◊ **HEAT TRANSFER & AIRCRAFT TRAJECTORY OPTIMIZATION**

Bachelor of Aerospace Engineering

Politecnico di Milano, Milan, Italy

12/09/2020-27/09/2023

UNIVERSITY PROJECTS

- ◊ **Master's Thesis (ongoing):** Experimental and numerical (CFD) investigation of Flettner rotors in the wind tunnel facilities of Politecnico di Milano, focusing on aerodynamic characterization of the Magnus effect across varying Reynolds numbers, velocity ratios and end-plate configurations.
- ◊ **Design of 2.3MW wind turbine:** multidisciplinary aero-servo-elastic design and optimization of an IEC Class IA wind turbine using Cp-Lambda, focusing on aeroelastic constraints and Cost of Energy minimization.
- ◊ **CFD Analysis of a Formula 1 front wing:** unsteady SU2 simulations analyzing ground effect and porpoising phenomena using compressible URANS formulations.
- ◊ **Experimental fluid dynamics labs:** wind tunnel testing on aerodynamic forces, flow behavior, and PIV analysis of vortex dynamics in side-by-side rotor eVTOL configurations.
- ◊ **Aerodynamic analysis of travel bus:** drag reduction of a long-distance coach through OpenFOAM CFD simulations and wind tunnel validation.
- ◊ **Flight testing - Pipistrel Virus SW121A:** experimental flight test campaign in collaboration with Pipistrel Italia, including test planning, data acquisition and post-flight analysis.
- ◊ **Crosswind rollover of a Heavy Truck:** dynamic stability analysis under transient crosswind conditions through aerodynamic modeling and multi-DOF vehicle simulation in MATLAB.

PROFESSIONAL WORK EXPERIENCE

Assistant aircraft designer - Internship - June and July 2019

- assisted employees in compiling maintenance records for the Superjet Sukhoi 100 aircraft
- participated in data archiving processes related to aircraft maintenance documentation

OTHER WORK EXPERIENCE

- ◊ **LIFEGUARD** - Turismo Srl- Jesolo, Italy - July and August 2018
- ◊ **AIR FOOD SRL** - Salesperson - San Siro Stadium, Milan, Italy 2022
- ◊ **WAITER** - Gattopardo, Milan, Italy - 2022